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CRUD: Read 7m

The read operation performs the **select** function, used to read data from a database.

To read data from the database, you must execute a select statement against the data. The MySQLi functions used in the next example for querying the database are:

- **`mysqli_query(mysqli $mysql, string $query, int $result_mode)` or `mysqli::query(string $query, int $result_mode)`** – performs a query against the database using the connection and query string passed to the function and returns a result set (the set of rows that match the query). The result mode has a default value.
- **`mysqli_num_rows(mysqli_result $result)` or `mysqli::$num_rows`** – returns the number of rows in the result set (passed as an argument). This is very useful when you want to iterate over the rows returned from the database query.
- **`mysqli_fetch_assoc(mysqli_result $result)` or `mysqli_result::fetch_assoc()`** – fetches the next row of data from the result set and returns it as an associative array.
- **`mysqli_fetch_row(mysqli_result $result)` or `mysqli_result::fetch_row()`** – like the above function, but fetches the next row of data from the result set and returns it as an enumerated array.
- **`mysqli_free_result(mysqli_result $result)` or `mysqli_result::free_result()`** – frees the memory associated with the result (to be used when you have finished processing the result set). This should always be executed when you have finished using the data. Very handy for long, complex scripts that might perform many database queries.

This example assumes you are querying the `tbl_restaurant` table from the restaurant database you created in the previous topic.
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If you need a refresher about this, we recommend taking a look at the tasks on the following page:

Topic 8 practice activity →

This example demonstrates setting up a query, sending it to the MySQL server and handling the results. Whenever you fetch a result, it is good practice to ensure it contains something. You will also notice you use a **while** loop to iterate through each row returned by the query.

The **mysqli_fetch_assoc** function accesses the record data using the field names in the array. The **mysqli_fetch_assoc** function makes for much easier handling and readability as you treat the data like any other associative array. The data is formatted in a fundamental way using HTML paragraph tags.

Avoid using **mysqli_fetch_all** unless the retrieved data will be small, because this function can consume a large amount of memory with badly written SQL or large tables.

Note that some of the HTML code gets echoed as strings as part of the PHP code.



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```
1  // prepare a query and send it to the server
2  $query = "SELECT rest_id, resta_name, resta_address, resta_category FROM tbl_restaurant";
3  $result = mysqli_query($db_connection, $query);
4
5  $row_count= mysqli_num_rows($result);
6  // check the result set for data
7  if($row_count > 0)
8  {
9
10 // show the number of rows (just as an example)
11     echo "<p>Record count: ".$row_count."</p>";
12
13 //iterate over the results and write them to the HTML page
14     $row = mysqli_fetch_assoc($result);
15     while ( $row)
16     {
17         echo "<p>Restaurant ID ".$row['rest_id']. "</p>";
18         echo "<p>Restaurant name ".$row['resta_name']. "</p>";
19         echo "<p>Restaurant address ".$row['resta_address']. "</p>";
20         echo "<p>Restaurant category ".$row['resta_category']. "</p>";
21         echo "<hr>";
22     }
23
24 // free any memory used by the query
25     mysqli_free_result($result);
26 }
```



Thoughts on this page?

